Zero-bias transport and extremal Bernoulli arrays

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Abstract

For a finite sum $S = \sum_{i=1}^{n} X_i$ of independent centered real random variables with total variance $V = \sum_{i=1}^{n} \mathbb{E}X_i^2 > 0$ and third absolute moment sum $R = \sum_{i=1}^{n} \mathbb{E}|X_i|^3 < \infty$, the L^1-Wasserstein distance between S and its zero-biased distribution S^* satisfies $2VW_1(S, S^*) \leq R$. We prove that equality holds if and only if all nonconstant summands X_i share a single centered two-point distribution. The single-summand characterization uses Choquet's theorem on the compact convex set of centered laws with variance at most one to represent any unit-variance law as a barycenter of three-point laws of variance equal to one, and shows that three-point laws and mixtures of distinct two-point laws have strictly smaller transport distance. Across multiple summands, unequal support diameters incur strict convolution loss, while for common-diameter Bernoulli arrays an exact conditional-variance identity expresses the transport deficit $R/(2V) - W_1(S, S^*)$ as a weighted variance of the success probabilities. We then establish a sharp inequality relating lattice span, skewness, and zero-bias transport, where equality holds precisely for centered binomial laws with Esseen parameter $p_{\mathrm{E}} = (4 - \sqrt{10})/2$ or $1 - p_{\mathrm{E}}$. Combining these two rigidity theorems determines the exact asymptotic Berry–Esseen ratio when a fixed finite block of independent summands is repeated independently.

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1 Introduction and main results

1.1 The zero-bias transport inequality and its equality cases

In Esseen's classical construction [3], the extremal constant $C_{\mathrm{E}} = (3+\sqrt{10})/(6\sqrt{2\pi})$ in the Berry–Esseen inequality is approached by sums of identically distributed two-point random variables. When one passes from identically distributed summands to general independent arrays $(X_1, \dots, X_n)$, two structural questions arise: what mechanism forces extremal summands to be supported on two points, and what forces distinct summands to share the same two-point distribution?

In this paper we answer both questions completely for the zero-bias Wasserstein transport inequality, and then combine that characterization with a sharp lattice transport inequality to determine the extremal arrays for repeated independent blocks.

For probability measures μ, ν on $\mathbb{R}$ with finite first moments, let

$$W_1(\mu,\nu) = \inf\{\mathbb{E}|U-V| : \mathcal{L}(U)=\mu,\ \mathcal{L}(V)=\nu\} = \int_{\mathbb{R}} |F_\mu(x) - F_\nu(x)|\,dx$$

denote their L^1-Wasserstein distance; we also write $W_1(U,V)$ for $W_1(\mathcal{L}(U),\mathcal{L}(V))$. If X is a centered real random variable with variance $v = \mathbb{E}X^2 > 0$, its zero-biased distribution X^*, introduced by Goldstein and Reinert [5, Definition 1.1 and Lemma 2.1], is characterized by the identity

$$\mathbb{E}[Xf(X)] = v\mathbb{E}f'(X^*) \tag{1.1}$$

for all absolutely continuous functions $f : \mathbb{R} \to \mathbb{R}$ for which both expectations exist. Goldstein [4, Lemma 1.1] and Tyurin [6, Theorem 3] independently proved the sharp single-variable transport inequality

$$W_1(X, X^*) \le \frac{\mathbb{E}|X|^3}{2v} \tag{1.2}$$

for every centered random variable with finite third absolute moment, and observed that every nonconstant two-point distribution attains equality in (1.2).

Now let $X_1, \dots, X_n$ be independent centered real random variables with

$$S = \sum_{i=1}^n X_i, \qquad V = \sum_{i=1}^n \mathbb{E}X_i^2 > 0, \qquad R = \sum_{i=1}^n \mathbb{E}|X_i|^3 < \infty. \tag{1.3}$$

Combining (1.2) with the independent-sum replacement characterization of S^* yields the additive zero-bias transport inequality $2VW_1(S,S^*) \le R$ [6, Corollary 1]. Our first theorem characterizes all equality cases in both (1.2) and its independent-sum extension.

Theorem 1.1 (Rigidity of zero-bias transport)**.** *For every finite independent array satisfying* (1.3)*,*

$$2VW_1(S,S^*) \le R. \tag{1.4}$$

Equality holds in (1.4) *if and only if all nonconstant summands X_i have one identical centered two-point distribution (and any remaining summands are identically zero). In particular, for $n = 1$, equality holds in* (1.2) *if and only if X has a two-point distribution.*

The distinction between the sum of individual third absolute moments $R = \sum_{i=1}^n \mathbb{E}|X_i|^3$ and the third absolute moment of the sum $\mathbb{E}|S|^3$ is essential in Theorem 1.1. When S is a sum of $n \ge 2$ independent, identically distributed nonconstant two-point variables, S is a centered binomial variable supported on $n+1 \ge 3$ points; by the $n = 1$ case of Theorem 1.1, its single-variable bound $2VW_1(S,S^*) < \mathbb{E}|S|^3$ is strict, whereas by the n-summand case of Theorem 1.1, its additive bound $2VW_1(S,S^*) = R$ holds with exact equality.

1.2 Lattice transport and repeated independent blocks

Our second main result applies Theorem 1.1 to the Berry–Esseen ratio of arrays formed by repeating a fixed finite block of independent summands. Let $S_1, S_2, \dots$ be independent copies of the block sum S in (1.3). For each integer $m \geq 1$, the combined array of mn summands has total variance mV and third absolute moment sum mR, so its Berry–Esseen ratio is

$$C_m = \frac{\sqrt{m}\, V^{3/2}}{R} \sup_{x \in \mathbb{R}} \left| \mathbb{P}\left\{ \frac{S_1 + \cdots + S_m}{\sqrt{mV}} \leq x \right\} - \Phi(x) \right|. \tag{1.5}$$

Set

$$C_{\mathrm{E}} = \frac{3 + \sqrt{10}}{6\sqrt{2\pi}}, \qquad p_{\mathrm{E}} = \frac{4 - \sqrt{10}}{2}.$$

Theorem 1.2 (Asymptotic Berry–Esseen ratio for repeated blocks)**.** *For any fixed independent block $(X_1, \dots, X_n)$ satisfying (1.3), the limit of (1.5) as $m \to \infty$ exists and satisfies*

$$\lim_{m \to \infty} C_m \leq C_{\mathrm{E}} \frac{2VW_1(S, S^*)}{R} \leq C_{\mathrm{E}}. \tag{1.6}$$

Equality $\lim_{m\to\infty} C_m = C_{\mathrm{E}}$ holds if and only if, for some $h > 0$, every nonconstant summand X_i has the distribution of $h(B - p_{\mathrm{E}})$ with $B \sim \mathrm{Bern}(p_{\mathrm{E}})$, or every nonconstant summand has the reflected distribution $h(p_{\mathrm{E}} - B)$, and any remaining summands are identically zero.

The bridge between Esseen's asymptotic expansion for $S_1 + \cdots + S_m$ and the zero-bias transport distance $W_1(S, S^*)$ in (1.6) is a sharp inequality for centered lattice distributions proved in Section 5 (Theorem 5.1): for any centered random variable X of variance $v > 0$ supported on a coset $a + h\mathbb{Z}$ ($h > 0$) with third moment $\kappa = \mathbb{E}X^3$,

$$|\kappa| + 3hv \leq (3 + \sqrt{10})\, 2vW_1(X, X^*),$$

with equality if and only if $X \overset{d}{=} h(N - np)$, where $N \sim \mathrm{Bin}(n, p)$ ($n \geq 1$) and $p \in \{p_{\mathrm{E}}, 1 - p_{\mathrm{E}}\}$.

Notice how the two inequalities in (1.6) play complementary roles:

(i) The first inequality in (1.6) depends only on the distribution of the block sum S, and attains equality whenever S is a centered binomial random variable $h(\mathrm{Bin}(k, p_{\mathrm{E}}) - kp_{\mathrm{E}})$ (or its reflection) for any $k \geq 1$.

(ii) The second inequality in (1.6) is Theorem 1.1, which compares $2VW_1(S, S^*)$ against the sum of the third absolute moments R of the individual summands $X_1, \dots, X_n$; it attains equality if and only if the binomial sum S is decomposed into identically distributed two-point summands $h(B_i - p_{\mathrm{E}})$ and zeros.

1.3 Open questions and overview of the proofs

Theorem 1.2 treats the limit $m \to \infty$ in which a fixed finite block $(X_1, \dots, X_n)$ is repeated independently. For general triangular arrays $(X_{n,1}, \dots, X_{n,k_n})$ whose laws may change with n, a companion paper [1, Theorem 1.1] proves that whenever the Lyapunov ratio $L_n = R_n / V_n^{3/2} \to 0$ and the Berry–Esseen ratio $C_n \to C_{\mathrm{E}}$, the normalized transport ratio $2V_n W_1(S_n, S_n^*)/R_n$ converges to 1, and that for actual finite-n global maximizers with sufficiently small L_n, Theorem 1.1's exact equality class is recovered. Determining whether C_n is attained by identically distributed two-point summands

and zeros at *every* finite n (without a small-Lyapunov or common-diameter restriction) remains open.

The remainder of the paper is organized as follows. Section 2 reviews the density and moment formulas for X^* and identifies the three inequalities in the independent-sum transport chain (2.5). Section 3 proves the single-summand characterization in Theorem 1.1 using a fixed-variance Choquet representation over three-point extreme laws. Section 4 completes the proof of Theorem 1.1 for n summands by proving strict convolution contraction for unequal diameters and establishing an exact conditional-variance identity (Proposition 4.1) for common-diameter Bernoulli arrays. Section 5 proves the lattice span–transport inequality (Theorem 5.1), and Section 6 combines it with Theorem 1.1 to prove Theorem 1.2.

2 Zero biasing and the independent-sum transport chain

We recall three basic properties of the zero-bias transformation from Goldstein and Reinert [5, Lemma 2.1]. First, for a centered random variable X of variance $v > 0$, the distribution of X^* is absolutely continuous with respect to Lebesgue measure, with density

$$d_X(u) = \frac{1}{v}\mathbb{E}[X\mathbf{1}_{\{X>u\}}] = -\frac{1}{v}\mathbb{E}[X\mathbf{1}_{\{X\le u\}}]. \tag{2.1}$$

Second, substituting $f(x) = x^2/2$ and $f(x) = x|x|/2$ into (1.1) gives the first moments

$$\mathbb{E}X^* = \frac{\mathbb{E}X^3}{2v}, \qquad \mathbb{E}|X^*| = \frac{\mathbb{E}|X|^3}{2v}. \tag{2.2}$$

Third, for any two probability laws μ, ν on $\mathbb{R}$ with finite first moments, the optimal quantile coupling [4, eqs. (15)–(16)] expresses W_1 as the L^1 distance between their distribution functions:

$$W_1(\mu,\nu) = \int_{\mathbb{R}} |F_\mu(x) - F_\nu(x)|\,dx. \tag{2.3}$$

Now consider the independent array (1.3). Omitting any summands of variance zero, write $v_i = \mathbb{E}X_i^2$, $\rho_i = \mathbb{E}|X_i|^3$, and $T_i = S - X_i$. Let $I \in \{1,\dots,n\}$ be a random index chosen independently of $(X_1,\dots,X_n)$ with $\mathbb{P}(I = i) = v_i/V$, and replace X_I by an independent random variable X_I^* having the zero-biased distribution of X_I. Conditioning on T_i in (1.1) gives $\mathbb{E}[Sf(S)] = \sum_{i=1}^n v_i\mathbb{E}f'(T_i + X_i^*)$, so that

$$S^* \overset{d}{=} T_I + X_I^*. \tag{2.4}$$

Applying the representation (2.3) and the triangle inequality to the mixture (2.4), followed by contraction of W_1 under convolution with T_i and the single-summand bound (1.2), gives the three-step inequality chain

$$\begin{aligned} W_1(S,S^*) &\le \sum_{i=1}^n \frac{v_i}{V} W_1(S, T_i + X_i^*) \\ &\le \sum_{i=1}^n \frac{v_i}{V} W_1(X_i, X_i^*) \le \frac{R}{2V}. \end{aligned} \tag{2.5}$$

Equality in (1.4) holds if and only if all three inequalities in (2.5) are equalities: every summand must attain equality in (1.2), convolution with T_i must preserve $W_1(X_i, X_i^*)$ for every i, and the signed CDF differences $F_{T_i+X_i} - F_{T_i+X_i^*}$ must have a common sign across i almost everywhere. The next two sections analyze these conditions in turn.

3 Equality for a single distribution

For a centered probability measure μ on $\mathbb{R}$ with variance $v > 0$ and third absolute moment $\rho = \int |x|^3 \, \mu(dx) < \infty$, define the scale- and reflection-invariant transport ratio

$$B(\mu) = \frac{2vW_1(\mu, \mu^*)}{\rho} \in [0, 1].$$

3.1 Two-point distributions and strictness for three-point distributions

Any centered two-point distribution takes the form $\mathcal{L}(h(B - p))$ with $h > 0$, $0 < p < 1$, $q = 1 - p$, and $B \sim \mathrm{Bern}(p)$. Its variance is $v = h^2pq$, its third absolute moment is $\rho = h^3pq(p^2 + q^2)$, and by (2.1) its zero-biased distribution μ^* is uniform on $[-hp, hq]$. Direct integration of $|F_\mu - F_{\mu^*}|$ on $[-hp, 0)$ and $[0, hq]$ gives

$$W_1(\mu, \mu^*) = \frac{h}{2}(p^2 + q^2) = \frac{\rho}{2v}, \tag{3.1}$$

so $B(\mu) = 1$ [4, Lemma 1.2].

Lemma 3.1. *Every centered distribution μ supported on three distinct points of positive mass satisfies $B(\mu) < 1$.*

Proof. First suppose $0 \notin \operatorname{supp} \mu$. After reflection if necessary, we may write $\operatorname{supp} \mu = \{x, y, z\}$ with $x < y < 0 < z$. Let μ_1 and μ_0 be the unique centered two-point distributions supported on $\{x, z\}$ and $\{y, z\}$, respectively. Then $\mu = \alpha\mu_1 + (1 - \alpha)\mu_0$ for a unique $\alpha \in (0, 1)$. Since μ_1 and μ_0 have variances $-xz$ and $-yz$, the zero-biased distribution μ^* is the variance-reweighted mixture [4, Theorem 2.1]

$$\mu^* = \beta\mu_1^* + (1 - \beta)\mu_0^*, \qquad \beta = \frac{\alpha x}{\alpha x + (1 - \alpha)y} > \alpha.$$

Writing $D_j = W_1(\mu_j, \mu_j^*)$ and $D_{10} = W_1(\mu_0, \mu_1^*)$, the triangle inequality gives

$$W_1(\mu, \mu^*) \le \alpha D_1 + (1 - \beta)D_0 + (\beta - \alpha)D_{10}, \tag{3.2}$$

whereas the two-point identity (3.1) for μ_1, μ_0 gives $\rho/(2v) = \beta D_1 + (1 - \beta)D_0$. Integrating the affine pieces of $F_{\mu_1^*} - F_{\mu_0}$ explicitly (compare Goldstein [4, Lemma 3.1 and eq. (28)]) yields

$$D_1 - D_{10} = \begin{cases} \dfrac{-y(y - x)\{y^2 + 2z^2 - y(x + 2z)\}}{(z - x)(z - y)^2}, & y(x + z) \le y^2 + z^2, \\[2ex] \dfrac{z^2}{z - x}, & y(x + z) > y^2 + z^2. \end{cases}$$

In the first case, $y < 0 < z$ and $y(x + z) \le y^2 + z^2$ imply $y^2 + 2z^2 - y(x + 2z) \ge z(z - y) > 0$, so in both cases $D_1 - D_{10} > 0$. Subtracting (3.2) from $\rho/(2v)$ therefore gives $\rho/(2v) - W_1(\mu, \mu^*) \ge (\beta - \alpha)(D_1 - D_{10}) > 0$.

Second, suppose $0 \in \operatorname{supp} \mu$. Write $\operatorname{supp} \mu = \{-ph, 0, qh\}$ with $h > 0$, $p + q = 1$, $0 < p \le q$ (after reflection), middle atom mass $1 - \tau \in (0, 1)$, and endpoint masses τq and τp. Here μ^* is uniform on $[-ph, qh]$, and direct integration of $|F_\mu - F_{\mu^*}|$ on $[-ph, 0)$ and $[0, qh]$ gives, with $r = p^2 + q^2$,

$$B(\mu) = \begin{cases} (q - p + 2\tau^2p^2)/r, & \tau q \ge p, \\ 1 + 2\tau^2 - 4\tau pq/r, & \tau q < p. \end{cases} \tag{3.3}$$

Hence $1 - B(\mu)$ equals $2p^2(1 - \tau^2)/r > 0$ in the first case and $2\tau(2pq/r - \tau) > 0$ in the second (since $2pq/r \ge p/q > \tau$), proving $B(\mu) < 1$ in all cases. □

3.2 Choquet representation at fixed unit variance

To reduce an arbitrary centered distribution μ with $v = 1$ and $\rho < \infty$ to laws supported on at most three points without changing the variance of the components, consider the convex set of probability measures

$$\mathcal{K} = \left\{ \nu : \nu \text{ is a probability measure on } \mathbb{R},\ \int_{\mathbb{R}} x\,\nu(dx) = 0,\ \int_{\mathbb{R}} x^2\,\nu(dx) \le 1 \right\}.$$

The second-moment bound $\int x^2\,\nu(dx) \le 1$ implies tightness and uniform integrability of x, so $\mathcal{K}$ is compact and metrizable in the weak topology induced by $C_b(\mathbb{R})$. By Choquet's theorem [2, pp. 333–337, Corollary 2 and Lemma 5], there exists a probability measure π supported on the set of extreme points of $\mathcal{K}$ with barycenter μ: that is, $\mu(A) = \int_{\mathcal{K}} \nu(A)\,\pi(d\nu)$ for every Borel set $A \subset \mathbb{R}$.

Every extreme point ν of $\mathcal{K}$ is supported on at most three points: if ν charged four disjoint Borel sets $A_1, \dots, A_4$, their four moment vectors $(\nu(A_j), \int_{A_j} x\,d\nu, \int_{A_j} x^2\,d\nu) \in \mathbb{R}^3$ would be linearly dependent, so a nonzero linear combination $g = \sum_{j=1}^4 c_j \mathbf{1}_{A_j}$ would give two distinct measures $(1 \pm \varepsilon g)\nu \in \mathcal{K}$ with average ν. Moreover, by Tonelli's theorem,

$$1 = \int_{\mathbb{R}} x^2\,\mu(dx) = \int_{\mathcal{K}} \left(\int_{\mathbb{R}} x^2\,\nu(dx) \right) \pi(d\nu), \qquad \rho(\mu) = \int_{\mathcal{K}} \rho(\nu)\,\pi(d\nu). \tag{3.4}$$

Because $\int x^2\,\nu(dx) \le 1$ for every $\nu \in \mathcal{K}$, the first identity in (3.4) forces $\int x^2\,\nu(dx) = 1$ for π-almost every ν. Consequently, π-almost every component ν is a centered probability law of unit variance, so its zero-biased law is given by the measurable kernel

$$\nu^*(A) = \int_{\mathbb{R}} x \left(\int_0^x \mathbf{1}_A(u)\,du \right) \nu(dx), \tag{3.5}$$

and Tonelli's theorem yields the unweighted mixture identity

$$\mu^* = \int_{\mathcal{K}} \nu^*\,\pi(d\nu). \tag{3.6}$$

(As noted in Goldstein [4, Theorem 2.2 and Remark 2.2], equal component variances are essential so that μ^* and μ share the same mixing measure π.) Convexity of W_1 under mixtures and (1.2) then give

$$W_1(\mu, \mu^*) \le \int_{\mathcal{K}} W_1(\nu, \nu^*)\,\pi(d\nu) \le \frac{1}{2} \int_{\mathcal{K}} \rho(\nu)\,\pi(d\nu) = \frac{\rho(\mu)}{2}. \tag{3.7}$$

If $B(\mu) = 1$, then both inequalities in (3.7) must be equalities, so $W_1(\nu, \nu^*) = \rho(\nu)/2$ for π-almost every ν. By Lemma 3.1, π-almost every component ν must be a unit-variance two-point distribution.

3.3 Cancellation between distinct two-point components

Every centered unit-variance two-point distribution is uniquely parameterized by $a > 0$ as

$$\nu_a = \frac{1}{1+a^2}\delta_{-a} + \frac{a^2}{1+a^2}\delta_{1/a},$$

with signed third moment $c(a) = \int x^3\,\nu_a(dx) = 1/a - a$, which is strictly decreasing from $+\infty$ to $-\infty$ on $(0, \infty)$. On $(-a, 1/a)$, the signed CDF difference between ν_a and $\nu_a^* = \mathrm{Unif}[-a, 1/a]$ is

$$f_a(x) = F_{\nu_a}(x) - F_{\nu_a^*}(x) = \frac{1}{1+a^2} - \frac{x+a}{a+1/a} \qquad (-a < x < 1/a), \tag{3.8}$$

and $f_a(x) = 0$ for $x \notin [-a, 1/a]$. Observe that $f_a(x) > 0$ on $(-a, c(a))$ and $f_a(x) < 0$ on $(c(a), 1/a)$. For any $0 < a < b$, we have $\max\{-a, c(b)\} < \min\{1/b, c(a)\}$, so on the nonempty open interval

$$I_{a,b} = (\max\{-a, c(b)\},\ \min\{1/b, c(a)\}), \tag{3.9}$$

we have $f_a(x) > 0$ and $f_b(x) < 0$. If the induced mixing measure π on $(0, \infty)$ had two distinct points $a < b$ in its support, then for every x in a compact subinterval of $I_{a,b}$ both $\mathbb{E}_\pi(f_a(x))_+$ and $\mathbb{E}_\pi(f_a(x))_-$ would be strictly positive, giving $\mathbb{E}_\pi|f_a(x)| - |\mathbb{E}_\pi f_a(x)| = 2\min\{\mathbb{E}_\pi(f_a(x))_+, \mathbb{E}_\pi(f_a(x))_-\} > 0$ and making the first inequality in (3.7) strict. Thus π is a point mass at a single $a > 0$, proving the single-variable case of Theorem 1.1.

4 Equality for independent arrays

Suppose now that $(X_1, \dots, X_n)$ attains equality in (1.4). By the last inequality in (2.5) and Section 3, every nonconstant summand must be a centered two-point variable:

$$X_i = h_i(B_i - p_i), \qquad h_i > 0, \quad 0 < p_i < 1,$$

with independent $B_i \sim \mathrm{Bern}(p_i)$.

4.1 Strict convolution loss for unequal support diameters

Suppose that $h = \max_i h_i$ and some nonconstant summand has diameter $0 < \delta < h$. Relabel so that X_1 has diameter h and lower-atom probability $q = 1 - p_1 \in (0, 1)$. In uncentered coordinates, $f(x) = F_{X_1 + hp_1}(x) - F_{X_1^* + hp_1}(x) = q - x/h$ on $(0, h)$ and 0 outside $[0, h]$. Because $T_1 = \sum_{j \geq 2} X_j$ contains a summand of diameter δ, the distribution of T_1 assigns positive probability to two atoms separated by δ. The two corresponding translates of f have opposite signs on a translate of the nonempty open interval

$$(\max\{qh, \delta\},\ \min\{qh + \delta, h\}),$$

so $|\mathbb{E}f(x - T_1)| < \mathbb{E}|f(x - T_1)|$ on a set of positive Lebesgue measure, making the second inequality $W_1(S, T_1 + X_1^*) \leq W_1(X_1, X_1^*)$ in (2.5) strict. Thus equality in (1.4) forces all nonconstant summands to share a common diameter $h > 0$.

4.2 Exact conditional-variance identity for a common diameter

Assume now that all n nonconstant summands have common support diameter $h > 0$. Put

$$K = \sum_{i=1}^n B_i, \qquad \mu = \sum_{i=1}^n p_i, \qquad w_i = \frac{h^2 p_i(1 - p_i)}{V}, \qquad c_{i,k} = \mathbb{P}\left\{\sum_{j \neq i} B_j = k\right\},$$

and for each cell $0 \leq k \leq n - 1$, define the cell weight, posterior index distribution, and cell mean

$$b_k = \sum_{i=1}^n w_i c_{i,k} > 0, \qquad \pi_{i,k} = \frac{w_i c_{i,k}}{b_k}, \qquad Q_k = \sum_{i=1}^n \pi_{i,k}(1 - p_i), \tag{4.1}$$

where $\sum_{k=0}^{n-1} b_k = 1$ and $\sum_{i=1}^n \pi_{i,k} = 1$.

Proposition 4.1 (Conditional-variance identity for common-diameter Bernoulli arrays)**.** *For any common-diameter Bernoulli array,*

$$\frac{R}{2V} - W_1(S, S^*) = h \sum_{k=0}^{n-1} b_k \operatorname{Var}_{\pi_k}(p_i) = \frac{h}{2} \sum_{k=0}^{n-1} \frac{\sum_{i,j=1}^n w_i w_j c_{i,k} c_{j,k}(p_i - p_j)^2}{b_k}. \tag{4.2}$$

Proof. On the kth lattice cell $[h(k-\mu), h(k+1-\mu)]$, write $x = h(k-\mu+s)$ with $s \in (0,1)$. Conditioning on $\sum_{j\neq i} B_j$ shows that only the event $\{\sum_{j\neq i} B_j = k\}$ contributes to $F_S(x) - F_{T_i+X_i^*}(x)$ on this cell, giving $F_S(x) - F_{T_i+X_i^*}(x) = c_{i,k}(1-p_i-s)$. Averaging with the variance weights w_i via (2.4) yields $F_S(x) - F_{S^*}(x) = b_k(Q_k - s)$, whose absolute value has integral on the kth cell equal to

$$hb_k \int_0^1 |Q_k - s|\, ds = \frac{hb_k}{2}\{Q_k^2 + (1-Q_k)^2\}.$$

On the other hand, (3.1) and $\sum_k c_{i,k} = 1$ give

$$\frac{R}{2V} = \frac{h}{2}\sum_{i=1}^{n} w_i\{p_i^2 + (1-p_i)^2\} = \frac{h}{2}\sum_{k=0}^{n-1} b_k \mathbb{E}_{\pi_k}\{p_i^2 + (1-p_i)^2\}.$$

Subtracting the two expressions and noting that $\mathbb{E}_{\pi_k}\{p_i^2+(1-p_i)^2\} - \{Q_k^2+(1-Q_k)^2\} = 2\,\mathrm{Var}_{\pi_k}(p_i)$ proves (4.2). □

At $k=0$, $c_{i,0} = \prod_{j\neq i}(1-p_j) > 0$ for every i, so $\pi_{i,0} > 0$ for all $1 \le i \le n$. Thus the right-hand side of (4.2) vanishes if and only if $p_1 = \cdots = p_n$, in which case $Q_k = 1-p$ for all k and

$$W_1(S, S^*) = \frac{h}{2}\{p^2 + (1-p)^2\} = \frac{R}{2V}. \tag{4.3}$$

This completes the proof of Theorem 1.1.

5 Lattice span, skewness, and zero-bias transport

Let X be a centered lattice random variable of variance $v > 0$ and finite third absolute moment, supported on a coset $a + h\mathbb{Z}$ ($h > 0$), with signed third moment $\kappa = \mathbb{E}X^3$.

Theorem 5.1 (Sharp lattice span–transport inequality)**.** *For every centered random variable X supported on $a + h\mathbb{Z}$ ($h > 0$) with variance $v > 0$ and finite third absolute moment,*

$$|\kappa| + 3hv \le (3 + \sqrt{10})\, 2vW_1(X, X^*). \tag{5.1}$$

Equality holds in (5.1) if and only if $X \overset{d}{=} h(N - np)$, where $N \sim \mathrm{Bin}(n,p)$ for some integer $n \ge 1$ and $p \in \{p_{\mathrm{E}}, 1-p_{\mathrm{E}}\}$.

5.1 Proof of the inequality by cellwise Jensen convexification

By (2.1), the density d_X of X^* is constant on each open lattice cell $(a+kh, a+(k+1)h)$; denote its value there by $d_k \ge 0$. On cells with $d_k = 0$, $F_X - F_{X^*} = 0$. On cells with $d_k > 0$, set

$$\omega_k = hd_k, \qquad r_k = \frac{F_X(a+kh) - F_{X^*}(a+kh)}{hd_k},$$

$$\ell(r) = \int_0^1 |r-s|\, ds = \begin{cases} r^2 - r + 1/2, & 0 \le r \le 1, \\ |r - 1/2|, & r \notin [0,1]. \end{cases}$$

Then $\sum_k \omega_k = 1$, and within the kth cell $F_X(a+(k+s)h) - F_{X^*}(a+(k+s)h) = hd_k(r_k - s)$, so integrating $|F_X - F_{X^*}|$ and $F_X - F_{X^*}$ over $\mathbb{R}$ and using $\int_{\mathbb{R}}(F_X - F_{X^*})\,dx = \mathbb{E}X^* - \mathbb{E}X = \kappa/(2v)$ gives

$$\frac{W_1(X, X^*)}{h} = \sum_k \omega_k \ell(r_k), \qquad \sum_k \omega_k \left(r_k - \frac{1}{2}\right) = \frac{\kappa}{2vh}. \tag{5.2}$$

Because $\ell(r)$ is convex on $\mathbb{R}$, Jensen's inequality applied to (5.2) with $t = |\kappa|/(vh) = \left|\sum_k \omega_k(2r_k-1)\right|$ yields

$$2vW_1(X, X^*) = 2vh\sum_k \omega_k \ell(r_k) \geq \begin{cases} vh(1+t^2)/2, & 0 \leq t \leq 1, \\ vht, & t \geq 1. \end{cases} \tag{5.3}$$

For $0 \leq t \leq 1$, the ratio $(|\kappa| + 3hv)/[vh(1+t^2)/2] = 2(3+t)/(1+t^2)$ attains its unique maximum $3+\sqrt{10}$ at $t = \sqrt{10}-3 < 1$; for $t \geq 1$, $(|\kappa| + 3hv)/(vht) = (3+t)/t \leq 4 < 3+\sqrt{10}$. This proves (5.1). (Note that combining (5.1) with $2vW_1(X, X^*) \leq \mathbb{E}|X|^3$ immediately recovers Esseen's moment inequality $|\kappa| + 3hv \leq (3+\sqrt{10})\mathbb{E}|X|^3$ [3, Lemma 1].)

5.2 Characterization of the binomial equality cases

Equality in (5.1) requires $t = |\kappa|/(vh) = \sqrt{10}-3 \in (0,1)$, where ℓ is strictly convex, so all cells with $\omega_k > 0$ must share the same ratio $r_k = r$ satisfying

$$0 < r < 1, \qquad |2r-1| = \sqrt{10}-3. \tag{5.4}$$

At any lattice point $x \in a + h\mathbb{Z}$ with atom mass $p_x = \mathbb{P}(X = x)$ and adjacent cell densities d_- (to its left) and d_+ (to its right), the jump relations $p_x = h\{rd_+ + (1-r)d_-\}$ and $v(d_- - d_+) = xp_x$ solve to

$$hd_- = p_x\left(1 + \frac{rhx}{v}\right), \qquad hd_+ = p_x\left(1 - \frac{(1-r)hx}{v}\right). \tag{5.5}$$

No interior lattice point between the support extremes can have $p_x = 0$, since across a zero-mass grid point $d_- = d_+ > 0$ and r_k would decrease by 1, contradicting constancy of r_k. Nonnegativity of $d_\pm$ in (5.5) confines the support to $[-v/(rh), v/((1-r)h)]$, with extreme atoms at $x_- = -v/(rh)$ and $x_+ = v/((1-r)h)$ separated by $x_+ - x_- = nh$ for some integer $n \geq 1$. Setting $p = 1-r \in \{p_{\mathrm{E}}, 1-p_{\mathrm{E}}\}$, we have $v = np(1-p)h^2$ and $x_j = h(j-np)$ for $0 \leq j \leq n$. Equating the two expressions in (5.5) for the density on the cell (x_j, x_{j+1}) gives the binomial recursion

$$\frac{p_{x_{j+1}}}{p_{x_j}} = \frac{(n-j)p}{(j+1)(1-p)},$$

which identifies $\mathcal{L}(X)$ as $h(\mathrm{Bin}(n,p) - np)$. Conversely, (4.3) verifies that every such law attains equality in (5.1), completing the proof of Theorem 5.1.

6 Proof of Theorem 1.2: repeated independent blocks

Let $S = \sum_{i=1}^n X_i$ be the sum of the fixed block in (1.3), with variance $V = \sum_{i=1}^n \mathbb{E}X_i^2$, third signed moment $\kappa = \mathbb{E}S^3 = \sum_{i=1}^n \mathbb{E}X_i^3$, and maximal lattice span $h \geq 0$ (where $h = 0$ if the law of S is nonlattice). By Esseen's asymptotic expansion for sums of independent copies of a fixed distribution with finite third absolute moment [3, pp. 160–162, eqs. (6)–(7)],

$$\lim_{m\to\infty} \sqrt{m}\, d_{\mathrm{K}}\left(\frac{S_1 + \cdots + S_m}{\sqrt{mV}},\, Z\right) = \frac{\phi(0)}{6V^{3/2}}(|\kappa| + 3hV). \tag{6.1}$$

Multiplying (6.1) by $V^{3/2}/R$ gives

$$\lim_{m\to\infty} C_m = \frac{\phi(0)}{6R}(|\kappa| + 3hV). \tag{6.2}$$

If $h > 0$, applying Theorem 5.1 to S and dividing by R gives the first inequality in (1.6), and Theorem 1.1 gives the second. If $h = 0$, then $|\kappa| = 2V|\mathbb{E}S^*| \le 2VW_1(S, S^*)$ by (2.2), so

$$\lim_{m\to\infty} C_m \le \frac{\phi(0)}{6}\frac{2VW_1(S,S^*)}{R} \le \frac{\phi(0)}{6} < C_{\mathrm{E}}.$$

Finally, if $\lim_{m\to\infty} C_m = C_{\mathrm{E}}$, then $h > 0$ and both inequalities in (1.6) must hold with equality. By Theorem 1.1, all $k \ge 1$ nonconstant summands X_i share a common centered two-point law $g(B - p)$ ($g > 0$, $0 < p < 1$), so $S \overset{d}{=} g(\mathrm{Bin}(k, p) - kp)$ has maximal lattice span $h = g$ and $\kappa/(Vg) = 1 - 2p$. Equality in Theorem 5.1 then forces $|1 - 2p| = \sqrt{10} - 3$, i.e., $p \in \{p_{\mathrm{E}}, 1 - p_{\mathrm{E}}\}$. Conversely, any such block attains equality in both steps of (1.6). This completes the proof of Theorem 1.2.

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